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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Semester GPA and CGPA Calculation System

Submitted by

DHANUSHYA.B

25BCS108

Under the Guidance of

ARAVINDH

Technical Trainer

Technical Department, KG College of Arts and Science

Academic Year: **2025 – 2026**

APRIL 2026

Semester: 2

DECLARATION

I hereby declare that the project entitled **"Semester GPA and CGPA Calculation System"** submitted in partial fulfillment of the requirements for the course **Java programming** is a **record of my original work** carried out under the guidance of **ARAVIDH**, Technical Trainer, Technical Department, KG College of Arts and Science. I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

DHANUSHYA.B

CERTIFICATE

This is to certify that the project entitled **"Semester GPA and CGPA Calculation System** is a **bona fide work** carried out by **DHANUSHYA1-BD, DepartmentofComputerScienceKG, CollegeofArts& Science**, in partial fulfillment of the requirements for the course **Java Programming**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: SARAVANAMAPATTI

Date:10.04.2026

Submitted for the Viva-Voce Examination held on 10.04.2026

Guide: **Head of the Department**(Technical Trainer, KGCAS) (Technical Head, KGCAS)

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ABSTRACT

The **Semester GPA and CGPA Calculation System** is a Java-based application designed to automate the calculation of students' academic performance. In many institutions, GPA and CGPA calculations are performed manually, which can lead to errors, delays, and inefficient record management. This project aims to simplify the calculation process by providing an automated system that computes GPA and CGPA accurately.

The system allows users to enter subject grades and credit values, and automatically calculates the GPA for each semester and the overall CGPA. The application is developed using Core Java programming concepts, including classes, objects, methods, conditional statements, loops, and exception handling. The project can be implemented either as a console-based application or using Java Swing for a graphical user interface.

The main objective of the system is to improve calculation efficiency, reduce human errors, and provide quick and accurate results. The system generates clear outputs showing subject-wise details, GPA, and overall CGPA.

This project demonstrates the practical implementation of object-oriented programming concepts in Java and provides a basic foundation for developing more advanced student management systems in the future.

1. INTRODUCTION

In the education system, accurate calculation of academic performance plays a crucial role in evaluating students and ensuring transparency. Traditionally, many institutions rely on manual methods to calculate Grade Point Average (GPA) and Cumulative Grade Point Average (CGPA), where calculations are done using paper or basic calculators. This manual process is time-consuming and can often lead to calculation errors, delays, and difficulty in maintaining proper academic records.

With the advancement of computer technology, automated calculation systems have become essential for improving efficiency and accuracy in academic management. A computerized system helps students and institutions quickly calculate GPA and CGPA, generate results automatically, and maintain organized academic records.

The **Semester GPA and CGPA Calculation System** is a Java-based application designed to automate the process of calculating students' academic performance. The system allows users to enter subject grades and credit values, and automatically calculates the GPA for each semester and the overall CGPA. The application generates clear and structured results displaying subject details, credits, GPA, and CGPA.

This project is developed using Core Java programming concepts such as classes, objects, methods, conditional statements, loops, and exception handling. The application can be implemented either as a console-based program or using Java Swing for a graphical user interface (GUI).

The primary goal of this system is to reduce manual effort, improve calculation accuracy, and provide quick results. This project also demonstrates how Java programming can be applied to develop simple real-world academic applications.

1.1 Objective

The main objective of the **Semester GPA and CGPA Calculation System** is to develop a simple and efficient application using Core Java that automates the process of calculating students' academic performance.

The specific objectives of this project are:

- To design and develop a **Java-based GPA and CGPA calculation system**
- To **automate the calculation** of GPA and CGPA using grades and credits
- To **reduce manual errors** in academic calculations
- To provide a **user-friendly system** for students to calculate results easily
- To demonstrate the implementation of **Core Java programming concepts** such as classes, objects, methods, loops, and conditional statements
- To generate **clear and accurate output** showing GPA and CGPA

1.2 Overview

The **Semester GPA and CGPA Calculation System** is a Java-based application developed to automate the calculation of students' academic performance. The system takes input such as subject grades and credit values, processes the data using standard GPA and CGPA formulas, and generates accurate results.

The application works in three main stages: **input, processing, and output**. In the input stage, the user enters subject-wise grades and corresponding credits. During processing, the system calculates the Semester GPA and overall CGPA using predefined formulas. In the output stage, the system displays the calculated GPA and CGPA in a clear and structured format.

This system is designed to be simple, efficient, and user-friendly. It can be implemented as a console-based application or enhanced with a graphical user interface using Java Swing. The system reduces manual effort, minimizes errors, and provides quick results, making it useful for both students and educational institutions.

1.3 Features

The **Semester GPA and CGPA Calculation System** provides a simple and efficient way to calculate students' academic performance. It is designed to reduce manual work and ensure accuracy by automatically processing grades and credit values. The system offers a user-friendly interface, quick result generation, and reliable performance. It supports multiple subjects and semesters, making it useful for tracking overall academic progress. Additionally, the system is flexible and can be enhanced into a complete student management application in the future.

Key Features:

- Automatic calculation of GPA and CGPA
- Easy input of grades and credit values
- Accurate and fast result generation
- Supports multiple subjects and semesters
- Reduces manual errors
- User-friendly interface (console/GUI)
- Clear and structured output display
- Lightweight and efficient system
- Based on Core Java and OOP concepts
- Scalable for future enhancements

1. PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the requirements of the system are identified. This stage helps in understanding how the system should work and what features are required to solve the problem efficiently.

In the **GPA and CGPA Calculation System**, the analysis phase focuses on identifying issues in the existing manual calculation process, such as errors, time consumption, and difficulty in handling multiple subjects and semesters. Based on this analysis, a computerized solution is proposed to automate the calculation process, **improve accuracy**, and provide quick and **reliable results.**

2.1 Problem Definition

In many educational institutions, the calculation of **GPA** and **CGPA** is performed manually using traditional methods such as paper-based calculations or basic calculators. This approach is time-consuming, prone to

human errors, and inefficient when dealing with multiple subjects and semesters. It also makes it difficult to maintain accurate academic records and delays the process of result generation.

Due to these limitations, students and staff may face difficulties in obtaining quick and reliable academic performance results. Therefore, there is a need for an automated system that can efficiently calculate GPA and CGPA, reduce errors, **save time**, and provide accurate and well-organized results.

2.2 Project Overview

The calculation of **GPA** and **CGPA** is an essential part of evaluating students' academic performance. In many institutions, this process is carried out manually, which can lead to inaccuracies and inefficiencies. As the number of subjects and semesters increases, the complexity of calculations also grows, making manual methods difficult to manage.

Manual systems often result in errors, consume more time, and lack proper organization of academic records. These issues can affect the accuracy of results and delay performance evaluation. Therefore, there is a need for an automated system that simplifies the calculation process, ensures accuracy, and provides quick and reliable results for students and institutions.

This project is implemented using **Core Java programming** concepts and can be developed as either a console-based application or a Java Swing **GUI** application. It utilizes object-oriented programming principles such as classes, objects, methods, loops, and conditional statements to efficiently calculate **GPA and CGPA.**

2.3 Scopes and Limitations

Scope

The **Semester GPA and CGPA Calculation System**focuses on automating and simplifying the process of evaluating students' academic performance. It provides a reliable and efficient way to calculate GPA for individual semesters and CGPA across multiple semesters using subject grades and credit values. The system ensures accuracy, reduces manual effort, and delivers quick results, making it useful for both students and academic institutions.\

The scope of this project includes handling multiple subjects with different credit weights, supporting calculations for several semesters, and displaying clear and structured results. It can be developed as a console-based application or extended into a graphical user interface using Java Swing. The system is designed to be simple, user-friendly, and efficient, making it suitable for basic academic use.

In addition, the project has the potential for future enhancements such as integrating a database for storing student records, generating detailed reports and transcripts, analyzing academic performance, and developing web or mobile-based applications. It can also be expanded into a complete student management system with features like attendance tracking, course management, and result analysis.

Scope Highlights:

- Automates GPA and CGPA calculation
- Supports multiple subjects and semesters
- Ensures accuracy and reduces errors
- Provides quick and clear results
- Can be implemented using Core Java (Console/GUI)
- Scalable for future enhancements like database and web integration

1. SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before the actual implementation. In this phase, the system is divided into different modules, and the workflow of the application is defined. Proper system design helps in organizing the program efficiently and makes development easier.

The **Semester GPA and CGPA Calculation System** is designed as a simple application where users can enter subject grades and credit values to calculate GPA and CGPA automatically.

The system is divided into several modules to perform specific tasks. Each module is responsible for handling a particular function of the system, such as input handling, calculation, and result display.

3.1 Modules

The main modules of the **Semester GPA and CGPA Calculation System** are explained below:

1. Menu Display Module

The **GPA andCGPACalculationSystem** is organized into multiple modules to ensure smooth and efficient operation. Each module is designed to handle a specific task, making the system structured, easy to understand, and simple to maintain. This modular approach improves the overall performance of the application and allows future enhancements without affecting the entire system.

Modules:

- **Main Display Module**
  This module provides a menu-driven interface for the user. It displays options such as entering subject details, calculating GPA, calculating CGPA, and exiting the program. It helps the user navigate through the system easily.
- **Subject Display Module**
  This module displays the list of subjects along with their credit values. It guides the user on what information needs to be entered for calculation.
- **Input Module**
  This module collects user input such as subject grades and credit values. It also ensures that the entered data is valid and stores it temporarily for processing.
- **Calculation Module**
  This is the core module of the system. It calculates grade points, computes the Semester GPA, and calculates the overall CGPA using standard formulas based on credits.
- **Result Display Module**
  This module presents the final output in a clear and structured format. It displays subject details, total credits, GPA, and CGPA for better understanding.
- **Validation Module**
  This module checks the correctness of user inputs, such as valid grade formats and appropriate credit values, to avoid errors during calculation.
- **Data Management Module**
  This module handles temporary storage and organization of academic data during program execution. It can be extended to support database storage in future.
- **Exit Module**
  This module allows the user to safely terminate the application after completing all operations.

1. METHODOLOGY

Methodology describes the systematic approach followed to develop the project. It outlines the steps taken from identifying the problem to designing, implementing, testing, and documenting the system. A well-defined methodology ensures that the system is developed in an organized manner and meets the required objectives efficiently.

For the **GPA and CGPA Calculation System**, a structured development process is followed to ensure accuracy, reliability, and ease of use. Each stage focuses on a specific aspect of development, helping in building a complete and functional application.

The development process includes the following stages:

- **Problem Analysis**
  Understanding the issues in manual GPA and CGPA calculation such as errors, time consumption, and difficulty in handling multiple subjects and semesters.
- **System Design**
  Dividing the system into modules like input, calculation, and result display, and defining the workflow of the application.
- **Program Development**
  Writing the Java program using Core Java concepts such as classes, objects, methods, loops, and conditional statements to implement the system logic.
- **Testing and Validation**
  Testing the system with different inputs to ensure correct GPA and CGPA calculations and validating the accuracy of results.
- **Result and Documentation**
  Displaying the final output clearly and preparing proper documentation explaining the working, features, and usage of the system.

4.1. Problem Analysis

In this phase, the problem is studied and analyzed in detail. The requirements of the system are identified based on the needs of academic performance calculation.

The main problems identified in manual GPA and CGPA calculation are:
• Manual calculations may lead to errors
• Calculating GPA and CGPA takes more time
• Difficult to handle multiple subjects and semesters efficiently
• Lack of proper record management for academic data

Based on these problems, the system requirements were defined to develop an automated application that simplifies GPA and CGPA calculation and improves accuracy and efficiency.

4.2. System Design

System design is the phase where the overall structure of the system is planned before implementation. It defines how the system will work, how different components are organized, and how data flows between them. A well-designed system ensures efficiency, accuracy, and ease of development.

In the **GPA and CGPA Calculation System**, the design is based on a simple and modular approach. The system is divided into different modules such as input, calculation, and result display. Each module performs a specific function, making the system easy to understand and maintain.

The workflow of the system follows an **Input → Processing → Output** model. The user enters subject grades and credit values as input. The system then processes this data by converting grades into grade points and applying formulas to calculate GPA and CGPA. Finally, the results are displayed in a clear and structured format.

The design can be implemented using Core Java concepts and can be developed either as a console-based application or with a graphical user interface using Java Swing. This structured design ensures that the system is efficient, user-friendly, and scalable for future enhancements.

4.3. Program Development

Program development is the phase where the actual coding of the system is carried out based on the system design. In this stage, the logic for calculating GPA and CGPA is implemented using a programming language.

In the **GPA and CGPA Calculation System**, the program is developed using Core Java. The application uses object-oriented programming concepts such as **classes**, **objects**, **methods**, **loops**, and conditional statements to perform calculations and manage data efficiently.

The development process includes writing code to accept user input (grades and credits), converting grades into grade points, and applying formulas to calculate GPA and CGPA. The program also includes validation to ensure correct input and displays the results in a clear and structured format.

The system can be implemented as a console-based application or enhanced with a graphical user interface using Java Swing. Proper coding practices are followed to ensure the program is simple, efficient, and easy to maintain.

4.4. Testing and Validation

Testing and validation is an important phase in software development where the developed system is checked to ensure that it works correctly and produces accurate results. This phase helps in identifying and fixing errors before the final deployment of the application.

In the **GPA and CGPA Calculation System**, the program is tested with different sets of input data such as various grades, credit values, and multiple semesters. This ensures that the system correctly calculates GPA and

CGPA under different conditions.

The system is validated by comparing the program's output with manually calculated results to verify accuracy. Input validation is also performed to ensure that only valid grades and credit values are accepted by the system.

Testing Activities Include:

- Checking correct input of grades and credits
- Verifying GPA calculation for a single semester
- Verifying CGPA calculation across multiple semesters
- Testing edge cases (minimum and maximum values)
- Ensuring proper display of results
- Handling invalid inputs (wrong grades, negative credits)

Through proper testing and validation, the system ensures **accuracy, reliability, and error-free performance**.

4.5. Result and Documentation

The **Result and Documentation** phase presents the final output of the system and provides a complete explanation of the project. After successful testing, the system generates accurate GPA and CGPA values based on the input grades and credit values.

The results are displayed in a clear and structured format, showing subject details, credit values, Semester GPA, and overall CGPA. The system ensures that the output is easy to understand and helps users quickly evaluate academic performance.

The documentation includes all details related to the project such as introduction, objectives, system design, methodology, modules, and results. It serves as a reference for understanding the working of the system and can be used for future enhancements or further development.

Result Features:

- Accurate calculation of GPA and CGPA
- Clear and organized result display
- Quick generation of results
- Easy interpretation of academic performance

Documentation Includes:

- Project overview and objectives
- System design and methodology
- Module descriptions
- Testing and validation details
- Final results and conclusion

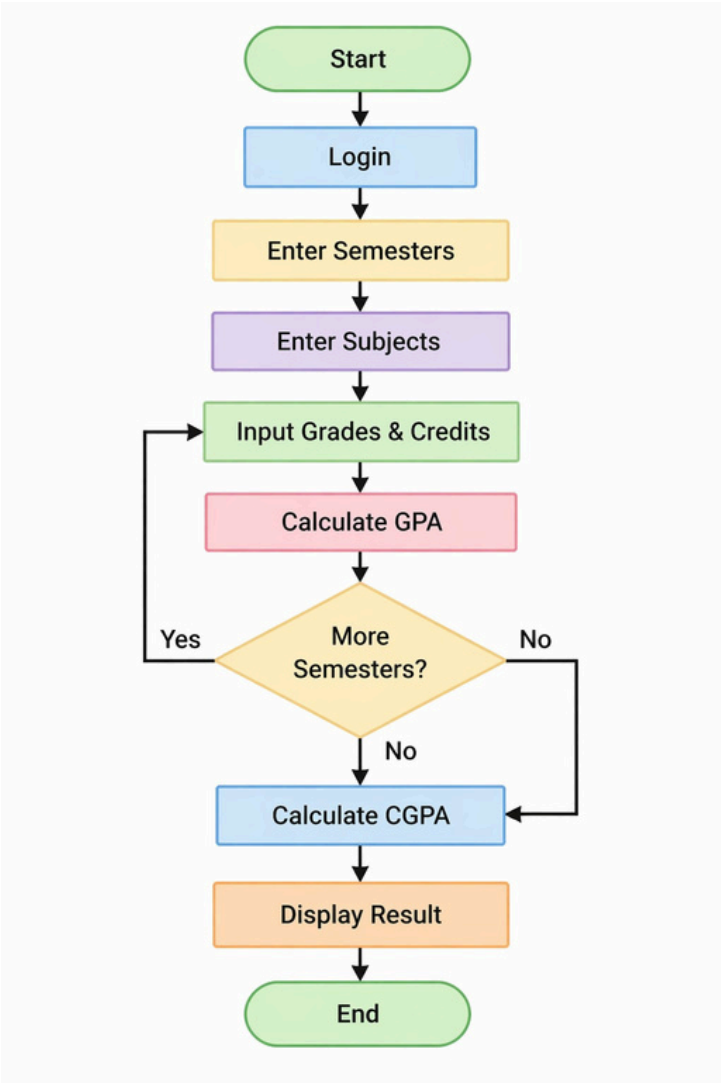
4.6. Algorithm

1. **Start the program**
2. Ask the user to **enter the number of semesters**
3. Initialize total variables(totalCredits=0,totalWeightedGPA = 0)
4. For each semester:
   - Ask the user to **enter number of subjects**
   - Initialize (semTotal = 0, semCredits = 0)
5. For each subject:

Enter **grade**
Enter **credit**
Convert grade to **grade point**
Calculate: gradePoint × credit
Add to semester total
Add credit to semester credits

6. Calculate **GPA = semTotal / semCredits**
7. Add GPA contribution to overall calculation
8. Repeatsteps forallsemesters
9. Calculate **CGPA = totalWeightedGPA / totalCredits**
10. **Display GPA and CGPA**
11. **End the program**

4.7. Flowchart



1. SOURCE CODE

```
import java.util.Scanner;

class GPACalculator04{



    public static String getPerformance(double gpa) {
        if(gpa >= 9) {
            return "Excellent";
        }else if (gpa >= 8) {
            return "Very Good";
        }else if (gpa >= 7) {
            return "Good";
        }else if (gpa >= 6) {
            return "Average";
        }else {
            return "Need Improvement";
        }
    }

    public static double[] calculateGPA(int subjects, Scanner sc) {
        double totalCredits = 0;
        double totalGradePoints = 0;

        for(int i = 1; i <= subjects; i++) {
            System.out.println("\nSubject " + i);
```

```
        System.out.print("Enter Credits: ");
        double credit = sc.nextDouble();

        System.out.print("Enter Grade Point: ");
        double grade = sc.nextDouble();

        totalCredits += credit;
        totalGradePoints += (credit * grade);
    }

    double gpa = totalGradePoints / totalCredits;

    System.out.println("---------------------------------");
    System.out.println("Semester GPA = " + String.format("%.2f", gpa));
    System.out.println("Performance: " + getPerformance(gpa));
    System.out.println("---------------------------------");

    return new double[]{totalCredits, totalGradePoints};
}

public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);

    System.out.println("==================================");
    System.out.println("     GPA & CGPA CALCULATOR");
    System.out.println("==================================");

    System.out.print("Enter Number of Semesters: ");
    int semesters = sc.nextInt();
```

```
        double overallCredits = 0;
        double overallGradePoints = 0;

        for(int i = 1; i <= semesters; i++) {
            System.out.println("\n******** SEMESTER " + i + " ********");

            System.out.print("Enter Number of Subjects: ");
            int subjects = sc.nextInt();

            double[] result = calculateGPA(subjects, sc);

            overallCredits += result[0];
            overallGradePoints += result[1];
        }

        double cgpa = overallGradePoints / overallCredits;

        System.out.println("\n==================================");
        System.out.println("        FINAL RESULT");
        System.out.println("==================================");
        System.out.println("Final CGPA = " + String.format("%.2f", cgpa));
        System.out.println("Overall Performance: " + getPerformance(cgpa));
        System.out.println("==================================");

        sc.close();
    }
}
```

1. DISCUSSION & CONCLUSION

7.1 Discussion

During the development of this project, various concepts of Java programming were applied, including:

- Loops (for handling multiple subjects and semesters)
- Methods (for modular and reusable code)
- Arrays (for handling multiple values)
- User input handling using Scanner

One of the key challenges was ensuring accurate CGPA calculation across multiple semesters. This was solved by using total credits and total grade points instead of averaging individual GPAs.

The project demonstrates how basic programming can be used to solve real-world academic problems effectively. Although the current system is console-based, it can be further improved by adding a graphical user interface (GUI), database storage, and grade conversion features.

7.2 Conclusion

The **GPA and CGPA Calculator**project has been successfully designed and implemented using Java programming.The system effectively calculates the Grade Point Average (GPA) for each semester and the overall Cumulative Grade Point Average (CGPA) based on user-provided inputs such as subject credits and grade points.

The project uses a **weighted average formula**, which ensures accurate and reliable results. By automating the calculation process, the system eliminates manual errors and reduces the time required for computing academic performance.

In addition, the application provides **performance analysis**, categorizing student results into levels such as Excellent, Very Good, Good, Average, and Needs Improvement. This feature helps users better understand their academic standing.

The project is simple, efficient, and user-friendly, making it suitable for students and educational purposes. It also serves as a strong foundation for learning core Java concepts and developing real-world applications.

1. REFERENCES

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